X-TREME PHOTO

PHOTO PREP

Staff User Guide

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Prep, Edit and Export for green-screen show photos

X-treme Photo Services
For staff use

Software by Ozark Mountain AI LLC
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New to Photo Prep? Read Quick start first, then keep this guide by the workstation. Problems go to the owner, with the photo and the error log (see Troubleshooting).

SECTION 01

About this guide

Photo Prep turns a night's green-screen photos into finished prints in three steps: **Prep**, **Edit**, **Export**. It replaces FXhome PhotoKey 6 Pro at X-treme Photo Services and works the same way where it can: tools on the left, the photo in the middle, a toolbox on the right.

This guide is the reference for staff. Read Quick start once, then come back to the section for whatever you are doing.

Two rules hold everywhere in the app:

- **Nothing is tied to a venue or a person.** There are no logins, no PINs and no venue lists. The app opens blank, and staff load the template for the show they are shooting.
- **Originals are never changed.** Prep writes copies into a separate folder. Edit and Export work from those copies. The photos off the camera card stay exactly as shot.

Photo Prep is developed by Ozark Mountain AI LLC and licensed to X-treme Photo Services. The version number is in the bottom-right corner of the window.

This guide is built into the app. Click **User Guide** at the bottom of the Photo Prep window, or press **F1**, to open it. It always matches the version installed.

SECTION 02

Door Mode, Meet and Greet or Group Photos

Photo Prep opens on a dashboard with three choices:

- **Door Mode** for photos from a memory card: **1 Prep → 2 Edit → 3 Export**. Most of this guide is about Door Mode.
- **Meet and Greet** for photos straight from the camera (Nikon D5300 over Wi-Fi, or USB): **1 Camera → 2 Edit**, with a **Print** button in Edit. See *Meet and Greet* below.
- **Group Photos** for big groups shot as they are, no green screen: **1 Prep** (load the photos and name each group) **→ 2 Edit → 3 Export**. See *Group Photos* below.

Photo folder (the strip under the three modes): click **Pick...** once and choose the photo backup folder, where each night's card folders are kept. From then on every "pick the photos" window (Prep's Browse, Group Photos' Browse, Edit's Import) opens there, or right beside the last card you picked inside it, so the other nights are one click away. **Open** shows it in File Explorer; **Change...** picks another. It's remembered on this PC. If the drive isn't plugged in, the strip says so and the windows open where they used to.

The last mode used on the PC is marked *Last used*. The mode button at the top, next to PHOTO PREP, takes you back to the dashboard to switch. Switching closes the photos that are open (nothing is deleted). Door Mode and Meet and Greet share the template; **Group Photos has its own templates**: switching into or

out of it starts with none loaded, a Door template won't open in Group Photos (nor a Group template in Door Mode), and a folder's remembered setup only comes back in the same kind of mode.

SECTION 03

Quick start: a normal night (Door Mode)

Most nights take five steps per card: open the show's template, Prep the card, check the AI's work, nudge what needs it, Export. Then Start New Card for the next one.

1. **Open the template.** Double-click the show's `.pptemplate` file, or in Edit click **Open...** next to the Template box. The backgrounds, overlay, caption and print size load with it.
2. **Prep the card.** Go to **1 Prep**, click **Browse...** next to *Pick the night's photos* and choose the card's folder. Tick **AI Optimize every photo when prep finishes**, then click **Start**.
3. **Check the AI's work.** When Prep finishes, click **Next: Edit**. Step through the photos with the **↓ / ↑ arrow keys** or the ▲ ▼ buttons under the preview. Photos the AI couldn't fix carry a **⚠ check** tag.
4. **Nudge what needs it.** Fix the tagged photos first. Whatever you change applies to the photo on screen only, so fixing one photo never touches the rest of the card.
5. **Export.** Go to **3 Export**, check the printer's hot folder, and click **Batch Export**. Tagged photos are held back until someone edits them.
6. **Next card.** Click **Start New Card** on the Export screen. You're back in Prep with the template still loaded; pick the next card's folder and click **Start**.

> **Tip.** no green screen at this account? Skip Prep. In Edit, click **Import** under the photo list instead. See *Importing without Prep* in the Edit section.

SECTION 04

The screen at a glance

The three steps are tabs at the top right; Edit, where most of the work happens, is laid out like PhotoKey.

AREA	WHERE	WHAT IT'S FOR
Step tabs	Top right: **1 Prep**, **2 Edit**, **3 Export**	Move between the three steps. You can go back and forth at any time.
Template bar	Edit, across the top	Template name box with **Open...**, **Save**, **New**; the view dropdown (**Full Color**, **Alpha**, **Foreground Original**); the sample dropdown (**Quarter**, **Half**, **Full Sample**); **Previous** / **Next** photo.
Tool list	Edit, left side	Canvas, Import, AI, Key, Masks, Filters, Position, Layers, Effects, Overlay, Text, Batch Export. Click one to show its settings in the Toolbox.

AREA	WHERE	WHAT IT'S FOR
Preview	Edit, centre	The finished print as it will export. Yellow boxes on it can be dragged.
Under the preview	Edit, below the photo	**Foreground** / **Background** tick boxes to hide either while you work, and ▲ ▼ to step through photos.
Toolbox	Edit, right side	The settings for the tool you picked, plus **Apply to all** and **Reset All** at the bottom.
Status line	Bottom of the Toolbox	Green = done, orange = needs a look. Read it after every action.
Footer	Very bottom	*Developed by Ozark Mountain AI LLC* (click for the About box) and the version number.

View modes. *Full Color* is the print. *Alpha* shows the cut-out in black and white (white = kept, black = keyed out): use it to spot holes or leftover green. *Foreground Original* shows the photo as shot.

Sample sizes. The preview opens on *Half Sample*: as sharp as the print at the size it's shown, and quick to update. *Quarter Sample* is faster but looks soft, so don't judge focus on it. *Full Sample* renders the whole print size for zooming in on fine detail; it's slower to update. Export always renders at full size.

SECTION 05

Step 1: Prep

Prep makes clean copies of the night's photos, ready to key: it fixes red eye and paints everything outside the green screen (walls, floor, light stands) the same green, so Edit only has one green to remove.

To run it:

1. Click **Browse...** next to *Pick the night's photos* and choose the folder from the card.
2. Leave the output folder as it is. Prepped copies go into a **Prepped** folder inside the input folder.
3. Tick the options you want (below) and click **Start**. **Stop** halts it; nothing already finished is lost.
4. When it's done, **Start** becomes **Next: Edit**, and the prepped photos open in Edit automatically.

OPTION	DEFAULT	WHAT IT DOES
Fix red eye	On	Finds faces and removes red from pupils. Only the pupils change, never skin.
Fill off-screen areas with green	On	Paints walls, floor and stands outside the curtain backdrop green. People, and anything they're holding, are left alone.
AI Optimize every photo when prep finishes	Off	Runs AI Optimize + Verify on every photo right after Prep, so the card comes out print-ready. Recommended.

OPTION	DEFAULT	WHAT IT DOES
Skip photos already in output	Off	Re-running Prep on the same folder only does new photos. Photos prepped by an older version of the person rules are redone automatically.
Watch input folder for new photos	Off	Keeps running and preps new photos as they land in the folder, e.g. straight off a tethered camera. Pairs with Edit's **Hot Folder**.
Save before/after previews	Off	Writes side-by-side previews into a `_previews` folder, for checking Prep's work.
Green sensitivity	25	Lower = more shades count as green. Leave it at 25 unless the log flags problems.

The log. Prep writes `_photo_prep_log.csv` into the Prepped folder. Rows marked **CHECK** are worth a human look: *no green backdrop, fill skipped* (the photo had no green screen and was left alone), *large fill* (an unusually big area was painted), or *could not process* (the file wouldn't open).

Prep checks its own work. Before painting anything green, Prep checks its plan with a second AI, the face finder used for red eye. Everyone has a body under their face, so if the plan would paint the area right under a face, Prep has made a mistake (a dark green shirt read as curtain, say). It does the same for legs and anything held below the waist, like a walker or a cane, when the detailed AI sees them standing in front of the curtain. It leaves that body as shot, tells Edit it's a person, and writes *CHECK - Prep corrected itself* in the log. The photo is fine; send it to the owner so the rules can be tuned.

Originals are never changed. Prep only ever writes into the Prepped folder.

SECTION 06

Step 2: Edit — the photo list

The photo list lives under **Import › Foreground**: one row per photo, with its tag. Everything you do in Edit happens on the photo selected there.

Moving through photos. Press the **↓ / ↑ arrow keys** (they work wherever you are in Edit, even right after moving a slider), click a row, use ▲ ▼ under the preview, or **Previous** / **Next** in the top bar. Page Down / Page Up still work too. Stepping stops at the first and last photo (it doesn't jump round to the other end), and hovering over ▲ or ▼ shows the photo on screen in a bar above the pointer (*DSC_0012.jpg · 3 of 40*). Type in **Search** to filter by file name; **Name ↑ / Name ↓** flips the sort.

Red rows. A photo flagged for any reason (⚠ check or ⚠ Prep check) shows as a **red row** with a red bar down its left edge, so it can't be missed while you scroll. It goes back to normal once someone edits that photo.

Tags on each row:

TAG	MEANING
⚠ check	The AI couldn't fix something. Held out of Export until someone edits the photo.

TAG	MEANING
⚠ Prep check	Prep flagged the photo when it painted the green (its note shows in **Masks**). Check it; editing the photo clears the flag.
edited	Someone changed this photo by hand.
AI ✓	AI Optimize ran and verified it clean.
as shot	Imported with no green screen; used as shot (see below).

Hot Folder. Tick **Use Hot Folder** and new photos from Prep (with *Watch input folder* on) appear in the list as they're prepped.

Remove takes the selected photo out of this batch. The original and its prepped copy stay on disk; prepping it again brings it back.

Importing without Prep

Import (under the photo list) brings photos straight into Edit, skipping Prep. Use it for accounts that don't shoot on green screen, or to add a few photos to the open batch.

Drag and drop works too: drag photos, or a whole folder, from File Explorer onto the photo list. Only .jpg, .png and .tif files come in; anything else in the folder is skipped (subfolders aren't opened). The Import button opens in the folder you last imported from.

- **No green screen:** the photo is used *as shot*. It fills the print and isn't keyed, and the row is tagged **as shot**. Red eye, filters, layers, overlay, text and AI Optimize all still work. The Key tool notes that its settings don't apply.
- **Green screen:** keyed as usual, without Prep's green fill.
- Red eye on imported photos is fixed live, so unticking *Red Eye* in the Key tool brings the original eyes back.
- If you later run the same photos through Prep, they're treated as normal prepped photos from then on.

Zooming in on the preview

Mouse wheel over the preview zooms in on the spot under the pointer (up to 8×), and out again down to the whole print. Zoomed in, **hold the left button and drag** to move the photo around (anywhere that isn't the yellow box or a mask point; a click still adds a mask point). The middle button drags too. **Ctrl+0** shows the whole print again. While zoomed, the preview is made at full size automatically, so it stays sharp; the toolbar shows how big it's shown. Zoom only changes the view, never the print.

Changes belong to one photo

Every change you make in Edit (key, filters, position, layers, effects, overlay size and spot, text, masks, red eye) applies to **the photo on screen only**. Photos you haven't touched show the template's look, so opening a template still reaches them.

- **Apply to All** puts this photo's look on every photo (each keeps its own position and masks). This is the one way to change the whole card at once.
- **Reset All** puts this photo back on the template's look. The small ↺ arrow on a section resets just that section.
- **Save Template** saves the template, not changes made on single photos. To save a photo's look as the template, click **Apply to All** first, then **Save**.

SECTION 07

Edit tools, one by one

Each tool on the left opens its settings in the Toolbox. Most have a tick box to switch the section on or off and a ↺ arrow to put it back to the template's setting. Yellow boxes on the preview can be dragged: inside to move, a corner to resize, the circle to rotate.

Canvas

- **Print Size:** 6 × 8 inch (default) or 8 × 10 inch, landscape, 300 dpi. Every photo exports at this size. Saved in the template. Meet and Greet always prints 8 × 10; the template's own size is kept for Door Mode.
- **Canvas Color:** tick **Use Color** to put a plain colour behind the people instead of a background image.
- **People Placement:** how Auto-fit and the AI place the group, in X-treme house style. The bottom of the print cuts people at the **Cut-off** (share of standing height), then the group is made as big as fits under the **Headroom** (space above the tallest head) and no wider than **Max width**, centred. Feet never show.

GROUP	HEADROOM	MAX WIDTH	CUT-OFF
Couples and trios	13%	85%	56% (mid-thigh)
Big groups (5+)	10%	96%	84% (mid-shin)

A group of 4 is cut halfway between the two. These numbers were fitted to X-treme's own finished prints; change them only if the house style changes. They're saved in the template.

If someone is cut off at the side or top of the original shot, the photo is scaled so the cut sits on the print's edge instead of floating inside it.

Import

- **Foreground:** the photo list (see *Step 2: Edit*).
- **Background: Import** adds background images to the library (or drag them onto the panel from File Explorer); click one to put it behind every photo. **Remove** takes it out. **Zoom & Crop** uses just part of the picture: **Zoom** up to 500%, then **Left / Right** and **Up / Down** (±100 puts that edge of the picture on the print's edge), or drag its yellow box.

Key

The Key removes the green screen. The defaults suit X-treme's curtains; most nights you won't touch it.

- **Red Eye:** on by default (fixed in Prep, or live on imported photos). Untick to keep this photo's eyes as shot. **Missed one?** Click **Select Eyes**, drag an oval round each red eye (as many as you like; right-click takes the last one back), then click **Remove Red Eye**. Only the red of the pupil changes, never the skin. **Undo Manual Fixes** removes them. Manual fixes belong to this photo only. **Esc** leaves selecting.
- **Auto Color Difference: Color** (Green or Blue screen), **Removal** (how hard green is removed), **Detail** (more keeps finer hair), **Softness** (how soft the edge is).
- **Matte: Clip Backg.** / **Clip Foreg.** tidy almost-clear and almost-solid edges; **Erode/Expand** shrinks or grows the cut-out edge.
- **Alpha Blur:** softens the cut-out edge.
- **Spill Suppression:** removes green light bounced onto people. **Spill Simulation** replaces it with the new background's light; **Desaturate** greys it out. Green clothing (a green shirt, a purse, lettering) keeps its colour.

The AI person-finder guides the key: deep inside a person is always kept and far away from everyone is always removed, so green clothes stay and wrinkles in the curtain go.

Masks

For what the key gets wrong, like a stool left in the shot or a hand that went missing.

Undo Prep's fill. Prep paints the floor and anything near people that isn't part of them green *before* Edit sees the photo, so a mask alone can't bring back something Prep painted over (a walker, a cane, a foot). At the top of Masks, **Undo Prep's Fill on This Photo** switches that one photo to the original as shot (red eye still fixed). Then mask by hand: **Retain Alpha** round what should stay, **Remove Alpha** round floor or stands that should go. **Put Prep's Fill Back** returns to Prep's version. It only ever affects the photo you're on. If Prep flagged the photo, its note shows here too.

1. Click **Add Mask**, then click points around the area. Each point joins the last, and points can go outside the photo. The points are small so they can sit close together round fine detail: zoom in (below) for hair, fingers or the edge of a sign.
2. Close the loop by clicking the first (yellow) point, or press **Enter** / click **Done**. Nothing is cut until the loop is closed.
3. Drag points to adjust.

Leave and come back any time. A mask you haven't closed keeps its points. Go to another tool or another photo, and when you come back it's marked *(not closed yet)* and picks up where you left off: keep clicking, then click the first point to close it.

Type: Remove Alpha (default) cuts the area out; **Retain Alpha** forces it to stay. **Feather** softens the mask edge; **Invert** flips it. Masks always belong to one photo. **Esc** stops drawing for now (the points stay).

Filters

Foreground filters act on the people; **Background** filters act on the background image.

- **Auto Brightness** and **Auto White Balance** are the AI's, measured on the people's faces.
- PhotoKey's full list: Transparency, White Balance (eyedropper), Auto Levels, Auto Contrast, Auto Color, Brightness, Contrast, Saturation, Levels, Color Temperature, Color Balance, Hue Colorize, Hue Shift, Sharpen, Blur, Bilateral Blur, Defocus, Diffuse, Vibrance, Leave Color, Half Tone, Half Tone Color, Silhouette, and Shadow Cast (people only).
- Tick a filter to switch it on. The arrows change the order they're applied in.

Position

- **Foreground (the people): Auto-fit** sizes and centres them on the floor line using People Placement. **Position** (Horizontal / Vertical), **Scale**, **Rotation** (Angle) and **Flip Horizontal**. Or use the yellow box on the photo.
- **Background:** the same **Zoom & Crop** as in Import.

Moving people moves them on this photo only.

Layers

For a prop, sign or logo between the background and the people, or in front of them.

- **Add Layer** and pick a PNG with transparency. A JPG works but shows as a box.
- **Size**, **Angle**, **Horizontal** / **Vertical**, or drag its yellow box. Tick **In front of the people** to put it on top. **Change image...** swaps the picture; **Remove Layer** deletes it.

Effects

- **Shadow Cast:** a soft shadow on the floor under the group. **Strength** sets how dark.
- **Light Wrap:** bleeds the background's light onto the edges of the people so they sit in the scene. **Amount** sets how much.

Overlay

The show's logo or frame on top of the print.

- **Load...** a PNG that's clear where the photo shows through. **Change...** swaps it; **Remove** takes it off.
- It starts in the bottom right. **Corner** switches between Bottom right and Bottom left. **Size**, **From side** and **From bottom** fine-tune it, or drag its yellow box (inside to move, a corner to resize).
- AI Optimize keeps it off faces and upper bodies (legs may run under a corner logo) and moves it to the other corner only if it has to.
- The overlay is saved inside the template.

Text

Add Text puts a caption on the print. `{date}` fills in the show date (from the photo's camera date). **Remove Text** takes it off.

In Group Photos, **Group Name** at the top of Text shows this photo's group name; fix a typo there and the corner label and the export name follow.

SECTION 08

AI Optimize + Verify

One click makes a photo ready to print: the AI fixes brightness, places the people, puts them on the background, keeps the overlay clear, then checks its own work and retries up to 3 times. What it decides becomes that photo's settings, so the preview shows exactly what will export.

To run it: open the **AI** tool. **Optimize Photo** does the photo on screen; **Optimize All Photos** does the whole batch. Or tick *AI Optimize every photo when prep finishes* on the Prep page. When it's done, **Go to Export** stays pinned at the bottom of the AI panel.

Each step can be ticked off:

STEP	WHAT IT DOES
Fix brightness	Brightens dark shots and darkens bright ones, measured on the people's faces. Also sets white balance.
Center the people	Sizes and centres the group the X-treme way (Canvas › People Placement). On an *as shot* photo, the photo is kept as shot instead.
Put the background behind them	Uses the background picked in Import.
Keep the overlay clear	Keeps it off faces and upper bodies; moves it to the other corner if it has to.
Verify, and redo until clean	Scans each finished photo, fixes what it finds (up to 3 tries) and flags anything still wrong for a person.

When a photo is flagged

A flagged photo gets the ⚠ **check** tag and its reason in the status line. It's held out of Export until someone edits it.

THE AI SAYS	WHAT TO DO
no people found in the photo	Check it's the right photo. If people are there, mask them in (Masks › Retain Alpha) or place them by hand.
no background loaded	Import › Background, pick one.

THE AI SAYS	WHAT TO DO
someone cut off at the ... edge of the original shot doesn't reach the print's edge	The camera cut someone off. Press **Auto-fit**, or drag the people so the cut sits on the print's edge.
people run off the ... edge of the print	Press **Auto-fit**, or make them smaller (Position › Scale).
the overlay covers someone	Move the overlay (drag its box or switch the Corner), or make it smaller.
a face is too dark / too bright	Filters › Brightness, or Levels.
green fringe left on the edges	Key › Spill Suppression at 100, or Matte › Erode/Expand a little negative.

After fixing by hand, the tag changes to **edited** and the photo goes into Export.

Re-running the AI on a photo replaces its position and brightness with the AI's choices; masks, layers and manual red-eye fixes are kept.

SECTION 09

Templates

A template is one show's whole setup in a single `.pptemplate` file. Templates replace venues: set a show up once, save it, and any PC can open it.

What a template holds: print size, People Placement numbers, backgrounds (the images are packed inside), the overlay (packed inside), caption, layers, effects, and the shared key and filter settings. It doesn't hold photos.

Build one:

1. Click **New** to start blank.
2. Set Canvas, load backgrounds (Import › Background), load the overlay, add a caption and any layers or effects. You can do all of this before any photos are open; the preview shows the template on its own.
3. Key, people filters, people position and masks need a photo, since they work on the people. Open a few test photos to tune them.
4. Type the show's name in the **Template** box and click **Save**. Templates save to *Documents › Photo Prep Templates* by default.

Open one:

- **Double-click** the `.pptemplate` file in File Explorer. Template files show the Photo Prep template icon. If Photo Prep is already open, the template loads into the open window.
- Or click **Open...** in Edit.
- Opening a template resets photo edits to the template's look, so it asks first if any photos have edits.

Share between PCs: the template is one file with every image inside it, so it works the same from Dropbox, a shared drive or a USB stick. Edit it on one PC, save, and copy the file to the others.

New clears everything back to blank. The app always opens blank; nothing is tied to a location or a person.

SECTION 10

Step 3: Export

Export renders every finished photo at full print size. By default the prints go into a **Ready to Print** folder, made next to the **Prepped** folder inside the night's photo folder:

```
Card folder (the night's photos, never changed)
├── Prepped          ← Prep's copies, what Edit works on
└── Ready to Print   ← finished prints from Export
```

1. Check the top lines: the **Template** in use, how many photos will export, and any held back.
2. Under **Save to**, leave **Save to a "Ready to Print" folder next to Prepped** ticked. To send prints somewhere else, like the printer's hot folder, untick it and pick the folder with **Browse...**. Photo Prep remembers your choice.
3. Click **Batch Export**. **Stop** halts it; photos already written stay.

The log ends with the night's total (*Tonight so far: 212 exported*), counted across every export this show night; it's written to the print log too. If the drive is getting full, Prep and Export say so before they start, so it can't run out halfway.

SETTING	DEFAULT	WHAT IT DOES
Print size	From Canvas	Shown for checking; change it in Edit › Canvas.
Format	JPG	JPG, TIFF or PNG.
Quality	100	JPG quality. Leave at 100 for prints.
Prefix / Suffix	Blank	Added before and after the file name. The *Example* line shows the result.
Embed color profile (sRGB)	On	Keeps colours consistent on the printer.
Keep the camera's photo info (EXIF)	On	Keeps the camera date and settings in the file.
Use Batched Backgrounds	Off	Exports every photo on **each** background in the library; the background's name is added to the file name.
Also save screen-size copies for a sales display	Off	Writes smaller copies for showing on a screen.
Save to a "Ready to Print" folder next to Prepped	On	Prints go in that night's photo folder. Untick to choose another folder.
Open the print folder when done	On	Opens the folder in File Explorer at the end.

Held-back photos. Photos tagged **⚠ check** don't export. The orange line under the count says how many. Fix or edit them in Edit, then click Batch Export again.

Start New Card. When a card is done, click **Start New Card** (above Batch Export). After you confirm, the photo list and the Prep and Export logs clear and Photo Prep goes back to **1 Prep** with an empty folder box, ready for the next card. The template stays loaded: backgrounds, overlay, canvas, text and look. Nothing is deleted: the finished card's prepped photos, edits and prints stay on disk, and opening its folder again brings it all back. Wait for an export to finish, and stop Prep if it's watching a folder, before starting a new card.

SECTION 11

Meet and Greet

For meet and greets: the camera sends each shot to the PC (over its own Wi-Fi or a USB cable), the shot comes straight into Edit already finished, and you print it on the spot. Entering Meet and Greet loads the AI in the background straight away, so the first customer doesn't wait for it. Meet and Greet always prints **8×10**, even with a 6×8 show template open. The template keeps its own size for Door Mode: switch back and it prints 6×8 again, and saving the template while in Meet and Greet saves it with its Door Mode size, not 8×10.

Step 1: Camera (Nikon D5300 over Wi-Fi or USB)

Pick how the shots come in at the top of the Camera page:

CHOICE	WHEN
D5300 over Wi-Fi	No cable: the PC joins the camera's own Wi-Fi (steps below).
D5300 over USB	The camera's USB cable plugged into the PC. Faster than Wi-Fi for full-size JPEGs, and nothing to join. Plug in, turn the camera on, close the Windows Photos import or Nikon Transfer if they pop up, and click **Connect to Camera**. Everything else (settings, test shot, the connection box) works the same as over Wi-Fi.
Camera software folder	Another program (digiCamControl) saves the shots into a folder and Photo Prep watches it.

Once per camera: on the D5300, **MENU › Setup › Wi-Fi › Network connection › Enable**. The camera makes its own Wi-Fi network, named like *Nikon_WU2_0090B5....*

Each event:

1. Turn the camera on with Wi-Fi enabled.
2. On the PC, join the camera's Wi-Fi network (Windows' Wi-Fi menu, bottom right). The connection box shows which Wi-Fi the PC is on, and warns if it isn't the camera's.

3. Open the show's template in Edit first if it isn't loaded, so each shot is finished with it.
4. On **1 Camera**, click **Connect to Camera**. That's it.

The connection box shows where things stand:

DOT	MEANS
Green: *Connected to D5300*	Ready. Battery level is shown, and a picture of the D5300 appears in the bottom-left corner of the Camera page. Every shot taken on the camera comes in by itself.
Amber: *Connecting…*	Looking for the camera.
Red: *Can't reach the camera / Connection lost – reconnecting*	The PC isn't on the camera's Wi-Fi, the camera's Wi-Fi went to sleep, or the signal dropped. Photo Prep keeps trying by itself; shots taken meanwhile come in once it's back.

The same dot shows next to **Camera** at the top of the window, so the printer can see it from Edit. Edit also has a **Photo Transfer** box in its lower-left corner: the connection, the shot coming over from the camera with a progress bar, and when the last one reached Edit with the count so far. Click it to go to the Camera step.

Each new shot is copied from the camera into the shots folder (a folder per show night, in *Pictures › Photo Prep Camera* unless you pick another), keyed (the green removed, nothing painted over), finished by the AI with the template's background, placement and brightness, and shown in Edit. Only shots taken after connecting come in, unless *Also bring in shots taken before connecting* is ticked. Edit follows the new shots while you're on the latest one; if you've gone back to fix an earlier photo (or are drawing a mask), you stay put and the new shot waits at the top of the list. **Print** waits until the AI has finished the shot you're on. Over Wi-Fi a full-size JPEG takes a few seconds to come across; set **Image size** to M and **Image quality** to JPEG normal for faster transfers (still plenty for an 8×10).

Changing the shots folder. Click **Browse…** under *Shots folder* to pick where shots are saved, connected or not. While connected, the camera stays connected: new shots go to the folder you picked (in a folder for the show night) and Edit switches to that folder's shots. The shots already taken stay in the old folder with their edits and prints. Use this to keep each show or group in its own folder.

Camera settings. Once connected, the **Camera settings** panel shows the camera's mode, shutter speed, aperture, ISO, exposure compensation, white balance, image quality, image size and battery. Pick a new value and it goes straight to the camera. Settings the mode dial controls are greyed out: on Auto or a scene mode the camera sets them itself, so turn the dial to **M**, **A**, **S** or **P** to change them from the PC. **Take a Test Shot** fires the shutter from the PC to check the settings.

The photographer keeps their own preview. Connected or not, the camera works as normal: every shot is saved to its memory card and shown on its screen. Photo Prep never starts live view, never takes over the shutter (except **Take a Test Shot**) and never deletes anything from the card; it only copies each shot after it's taken. If another program left the camera set to save shots to a computer instead of its card (digiCamControl can, which is why the preview disappears there), Photo Prep puts it back to the card when it connects and says so in the log. If shots still don't show on the camera's screen, check the camera's **Playback menu › Image review** is **On**.

Disconnect stops the connection. Shots already in stay in Edit; connecting again picks up where you left off.

If it won't connect. The red box says which step the camera stopped at. *Can't reach the camera* with "is this PC on the camera's Wi-Fi?" means the PC isn't on the camera's network (or the camera's Wi-Fi is off or asleep). *It stopped answering while…* means the PC reached the camera but it went quiet: turn the camera's Wi-Fi off and on (MENU › Setup › Wi-Fi › Network connection), close anything else that might be connected to it (Nikon's phone app, digiCamControl), and click **Connect to Camera** again. If it still won't connect, click **Connection log** and send that file to the owner: it records every step with the camera.

Show nights and folders. A show night runs until 5 AM: a show that goes past midnight, or a reconnect at 12:30, stays in the same folder. **Drive space** is checked when you connect and as shots come in; a warning shows in the log (and in the Photo Transfer box when it's nearly full) before it can run out mid-show.

If Photo Prep closes unexpectedly (a crash or a power cut) while Meet and Greet is connected, the next time it opens it asks to carry on: one click brings back the template, Meet and Greet, the camera connection and the same shots folder, and brings in the shots taken while it was down (not ones from earlier nights still on the card).

Camera software folder instead? Pick **Camera software folder**, point it at the folder digiCamControl (free, digicamcontrol.com) saves into, and click **Start Watching**. Camera settings are changed in digiCamControl in that case.

Step 2: Edit and Print

Edit works exactly as in Door Mode; the newest shot is at the top of the list and selected. Fix anything that needs it, then click **Print** at the top right (or press **Ctrl+P**). Photo Prep makes the full-size print and opens the **Print Preview**: the print on the chosen paper exactly as it will come out. Pick the **Printer**, **Paper**, **Orientation** (Automatic matches the photo) and **Copies**; **Printer Settings…** opens the full Windows print options (the printer's own preferences, such as borderless). The note under them says whether the print fills the paper edge to edge, or warns about white edges (paper that doesn't match the print) or a margin (turn on borderless). Click **Print** to send it, or **Cancel**. The printer and paper are remembered for next time. A flagged photo asks before printing. Leave **Orientation** on **Automatic**: it lays the print the right way round whichever way the printer's driver lists the paper (some list 8×10 as "10 × 8").

Smaller prints. Every printer remembers its own paper. For a 5×7 or 4×6 off the 8×10 printer, pick it under **Print size**. Photo Prep picks the smallest paper the printer offers that holds it, centres the print on it with corner marks to cut along, and the canvas doesn't change. **Scale to fit** (on by default) keeps the whole photo, with thin white edges because an 8×10 is a slightly different shape from a 5×7. Untick it to fill the 5×7 exactly; that trims a little off the photo's edges, so check that nothing important (text, the overlay) sits right at the edge.

Two on one sheet. *Two 5 × 7 on one sheet* puts two 5×7s side by side on one 8×10 sheet (half the media per 5×7), and *Two 4 × 6 on one sheet* puts two 4×6s on a 6×8 sheet; cut along the marks (or in half). Both are the same photo, so set Copies for how many pairs. Print size goes back to Full paper for the next print.

A preview that looks wrong on one of the printers: send the owner the file *printer-papers.txt*: type `%LOCALAPPDATA%\Photo Prep` in File Explorer's address bar to find it. It lists every paper that printer's driver offers, exactly as the driver describes it.

Reprint. Reprint... next to Print lists tonight's prints, newest first, with a look at each. Pick one and it goes through the Print Preview again from the same print file, at the same size (a 5×7, or two on one sheet), so it comes out exactly as before.

The Photo Transfer box (bottom left in Edit) also shows the printer and what Windows says about it: *Ready*, *Printing* with how many are waiting, or a problem in orange (*Offline*, *Out of paper or ribbon*, *Paper jam*, *Needs attention*). Under it, **Printed tonight** counts the prints made this show night.

The print log. Every print (and every Door Mode export) is written to *print-log.csv* next to the other logs: the show night, time, photo, size, copies, prints and printer. It opens in Excel for the end-of-night count. If it's open in Excel during the show, prints are still counted (and Reprint still works); they're written to the file once it's closed.

Where things go, all inside the shots folder:

FOLDER	WHAT'S IN IT
(the shots folder)	The shots as they came off the camera. Never changed.
Meet and Greet	Edit's copies and their settings.
Printed	Every print file sent to the printer, at 300 dpi.

SECTION 12

Group Photos

For big groups shot as they are: no green screen, no green fill, no red eye. Each photo goes into the template's frame (the show's art with a window cut for the group), gets auto color and brightness, and has the group's name printed under the title.

Step 1: Prep (load and name)

1. Click **Browse...** and pick the folder of group photos, or drag the folder (or just some photos) onto the window. Copies go in a **Prepped** folder inside it; the originals are never changed.
2. Every photo is listed with its thumbnail. Type the group's name in the box under it. **Enter** (or ↓) goes to the next box, ↑ to the one before.
3. Click **Next: Edit**.

Loading the same folder again only adds photos that are new; names already typed stay. A photo with no name keeps its file name and prints with no label.

Step 2: Edit

Open the template for the show. Each photo is placed in the frame's window automatically, the way X-treme's staff place them by hand: as wide as the print, zoomed in for a smaller group (never past the photo's own pixels), centred on the faces, with the front row at the same height every time. The frame trims

what's outside the window. Auto color and auto brightness are on by default (Filters); there's nothing to key. The photo list shows each group's name. To fix a name, use **Text › Group Name**. To nudge one photo, use **Position** as usual.

Clay Cooper or King's Castle

The two theatres are shot differently, and the templates know it: open the show's template and everything below comes with it. Nothing else to pick.

	CLAY COOPER	KING'S CASTLE
Group width (faces fill this much of the window)	88%	92%
Front row (lowest faces, down the window)	64%	47%
Max zoom for a small group	107%	145%
Theatre seats along the bottom	—	painted over with carpet
Group name	bold white, under the title	red or black serif, a word per line, beside the year

These were measured on X-treme's own finished prints. They're under **Canvas › Group Placement** (*Like* sets all three at once) and saved in the template.

Theatre seats (King's Castle). Photos taken from the house show the front-row seats along the bottom. **Masks › Theatre Seats** paints carpet over them: the stage's front edge is found from the carpet, the seats below it (and up the sides) are replaced with carpet in the same colour and texture. People are never painted over. It takes a few seconds per photo the first time. Turn it on in the template (Apply to All, then Save) so every photo gets it; untick it on a photo that wasn't taken from the house.

Group name style. Text › Group Name › Style: *Bold white, black outline* (Clay Cooper), *Red serif* or *Black serif* (King's Castle: red on Christmas Wonderland and Spirit of America, black on the Irish Tenors' orange). The serif styles put a word per line, up to three lines (*Cullman / Senior / Center*); more words are shared out over three. A long name shrinks to fit **Max width**.

Making a Group Photos template from a PSD

1. Start **Group Photos**, load a few photos, go to **Edit** and click **New** (Template) for a blank template.
2. **Canvas:** pick the print size (6×8 for the Clay Cooper frames).
3. **Overlay › Load:** pick the show's **.psd** (or a PNG). Photo Prep uses the picture as Photoshop shows it, transparency included; hidden layers don't show. The see-through window in the middle is where the group goes: Photo Prep finds it on its own, words inside it and all.
4. **Text › Group Name:** if the name runs into the art, move it with **Across** / **Down** and set its **Size**. Then click **Apply to All**.
5. **Save** the template with the show's name.

The PSD must be RGB, 8 or 16 bit, saved with *Maximize Compatibility* on (Photoshop's default). If Photo Prep says the file can't be read, open it in Photoshop and save it again with that setting.

A frame from any picture (Oval Window)

No PSD? Any full-size picture works: a flag, a starry background, a photo of the stage. In Group Photos, load it with **Overlay › Load**. A picture with no see-through parts gets an oval window cut out of it automatically. Under **Oval Window**:

- **Shape like** sets the window exactly as cut in one of the Clay Cooper PSDs: *Clay Cooper Theatre* (with its thin black outline), *Christmas*, *Hot Rods* (soft edge), *Matt Gumm*, *60s*.
- **Width**, **Height**, **Across**, **Down** fine-tune it; **Soft edge** fades the edge; **Outline** draws a black line round it.
- Untick **Cut an oval window** to use the picture as loaded.

The group goes in the window just as with a PSD. Add the show's title with the Text tool, or design it into the picture. Save the template when it looks right.

Step 3: Export

Same as Door Mode, except no background is needed. Each print file is named for its group: *Smith Family Reunion.jpg*. Two photos of the same group become *Smith Family Reunion.jpg* and *Smith Family Reunion_2.jpg*.

SECTION 13

Installing, updating and uninstalling

Photo Prep needs Windows 10 or 11, 64-bit. It installs for the signed-in Windows user, so no administrator password is needed.

JOB	HOW
New PC	Run **Photo Prep [version] Setup.exe** once (about a minute). It adds a *Photo Prep* icon to the desktop and Start menu; use that icon from then on.
Update	Automatic from version 2.3.9 on (below). By hand if ever needed: close Photo Prep, then run **Photo Prep [version] update.exe** once. It keeps Python and the AI models already on the PC and only swaps in the new app (a few seconds).
Uninstall	Settings › Apps › Installed apps › Photo Prep › Uninstall. It removes the app and shortcuts, asks before deleting saved templates, and never touches photos.

Automatic updates

Photo Prep updates itself. Staff don't need to do anything.

1. **When you open Photo Prep**, it takes a quick look for a newer version first. If there is one, an **Updating Photo Prep** box shows the download bar, then *Installing...*, and Photo Prep opens on the new version with a **You're updated** window showing what's new. Click **OK** and carry on. It takes under a minute; don't switch the PC off while the box is up. No internet? Photo Prep just opens as normal.
2. **While Photo Prep is open**, it checks again every 4 hours. A newer version found then downloads in the background, and the footer shows **Update [version] ready - installs when you close Photo Prep**. Nothing installs during Prep, AI or Export.
3. Close Photo Prep when you're finished. The update installs in a few seconds and Photo Prep reopens by itself with the **You're updated** window. Close it again if you're done for the night (it won't reopen a second time).
4. Every download is checked against its fingerprint, so a damaged download is thrown away and Photo Prep opens on the version it already had.
5. If a new version fails to open three times in a row, the previous version is put back automatically and that update is skipped.

FOOTER SAYS	MEANING
Update [version] ready - installs when you close Photo Prep	Nothing to do. Close Photo Prep when you're finished.
Updated to [version]	The update installed.
Update [version] was blocked by Windows Security	Windows quarantined the download. Follow *Windows security warnings* below, then click the footer message to check again.
Photo Prep [version] is out and needs the full installer	That release changes the AI models or Python. Tell the owner; it needs the full Setup.exe.

No internet? Photo Prep works normally and checks again later.

After an update to the person rules, Prep redoes previously prepped batches the next time it runs on them, even with *Skip photos already in output* ticked. That's deliberate.

Windows security warnings

Photo Prep isn't digitally signed yet, so Windows may warn about it. These warnings are false alarms.

- **"Windows protected your PC"** (blue box): click **More info**, then **Run anyway**.
- **Windows Security quarantines it** (*Trojan:Win32/Wacatac...!ml*): the `!ml` means Defender's machine-learning guess, not a known virus. Open **Windows Security › Virus & threat protection › Protection history**, click the entry, then **Actions › Restore** (or **Allow on device**).
- To stop it recurring on a PC: **Virus & threat protection › Manage settings › Exclusions › Add an exclusion › Folder**, and add the folder you keep the installers in plus `%LOCALAPPDATA%\Photo Prep`.

> Only do this for files that came from the owner. If a warning names a specific threat without `!ml`, don't restore it; tell the owner.

SECTION 14

Troubleshooting and FAQ

PROBLEM	FIX
Green showing between arms or legs	Switch the view to **Alpha** to see it. Mask it out (Masks › Remove Alpha) and tell the owner which photo; it helps tune the AI.
Part of someone is missing (a dark shirt, a hand)	Mask it back in with **Retain Alpha**, and report the photo.
A sign, poster or light touching someone's head stays in the photo	The AI reads anything touching a head as part of that person. Draw round it with **Masks › Remove Alpha**. Better still, keep guests a step away from wall signs at the edge of the curtain.
Green tint on hair or edges	Key › Spill Suppression at 100; or Matte › Erode/Expand slightly negative.
A photo won't export	It's tagged **⚠ check**. Read its reason, fix it, and export again.
Preview looks soft	You're probably on Quarter Sample. Switch back to **Half Sample** (the default); Full Sample for fine detail. The export is always full size.
Photo with no green screen	Import it from Edit (it comes in *as shot*) instead of running Prep.
Red eye missed	Key › Red Eye › **Select Eyes**, drag ovals, **Remove Red Eye**. Red eye is found automatically only when the whole face is in the shot.
"Photo Prep is already open"	It's running already; check the taskbar.
Something went wrong and the app restarted	Your template comes back automatically; reopen the batch from Prep's output folder if needed.
Windows Security deleted it	See *Windows security warnings* above.

Does the preview show the AI's changes, or are they only applied on export? The preview shows them. Everything the AI decides is saved as that photo's settings, and Export renders the same settings at full size.

Does Photo Prep change my original photos? Never. Prep writes copies, and Edit and Export work from those.

Where's the error log? Click **Send logs** at the bottom of the window: every log Photo Prep keeps (errors, the camera connection, the printers, the print log, settings, and a note about this PC) is zipped onto the Desktop, ready to attach to an email to the owner. Nothing is sent anywhere by itself. The logs live in `%LOCALAPPDATA%\Photo Prep`.

SECTION 15

Keyboard shortcuts and glossary

KEYS	DOES
F1	Open this user guide
↓ / ↑ (arrow keys)	Next / previous photo, from anywhere in Edit (not while drawing a mask or selecting eyes: finish or press Esc first). Page Down / Page Up do the same.
← / → on a slider	Nudge that slider by one step
Ctrl+P	Meet and Greet: print the photo on screen
Ctrl-click / Shift-click	Select several photos
Enter	Close the mask you're drawing
Esc	Stop drawing a mask, stop selecting eyes, or cancel the eyedropper
Right-click (while selecting eyes)	Take back the last oval

TERM	MEANING
Door Mode	Photos from a memory card: Prep, Edit, Export.
Meet and Greet	Photos straight from the camera (Wi-Fi or USB): Camera, Edit, Print.
digiCamControl	Free Windows software that saves each shot from a USB camera into a folder (Meet and Greet's *Camera software folder* choice).
Prep	Step 1: makes clean copies with red eye fixed and off-screen areas painted green.
Prepped folder	Where Prep's copies go, inside the card's folder.
Key / keying	Removing the green screen so the background can go behind people.
Alpha	The cut-out: white is kept, black is removed.
Spill	Green light from the screen bounced onto people's edges and hair.
As shot	A photo imported with no green screen; used whole, not keyed.
Template	One show's setup saved as a `.pptemplate` file.
Overlay	The show's logo or frame PNG on top of every print.
Layer	An extra PNG (a prop or sign) placed behind or in front of the people.
Ready to Print	The folder Export saves prints into by default, next to Prepped.
Hot folder	The folder a printer watches; Export can write there instead. Edit's *Use Hot Folder* is a different thing: it shows photos as Prep finishes them.
⚠ check	A photo the AI couldn't fix; held out of Export until edited.